



[BIT703 Web Technologies](#) / [Contents](#) / [Topic 12: Web application performance](#) / [Applied testing](#)

Performance testing types 45m

There are different types of performance tests, each with a specific objective. A tester should choose the type of test that's relevant to their application and enables them to confirm that the business objectives will be achieved.

Select each icon to find out about the different performance test types.



ENTER FULLSCREEN

[Skip to main content](#)

Types of performance testing

Load testing

Scalability testing

Soak testing

Stress testing

Spike testing

Failover testing

Volume testing



Load testing

Tests the performance of the system (such as response times and stability) under normal, daily workload (users or transactions) and peak workloads (such as the Boxing day load for retail management systems).



Performance testing tools

Performance testing may involve generating thousands of users or requests in a given amount of time. Therefore, it's not practical to conduct this type of testing manually. You'll usually need a load generation tool to generate the desired amount of load on the system.

Most load generation tools have monitoring capability so that they can capture the performance results during the test.

Different performance testing tools support different technology. Therefore, it is worth becoming familiar with a few performance testing tools and their capabilities, advantages and disadvantages.

Common performance test tools are:

- BlazeMeter
- LoadNinja
- JMeter
- NeoLoad
- ReadyAPI
- OpenText Professional Performance Engineering (LoadRunner Professional)
- StresStimulus
- LoadView
- WebLOAD

Load testing

The type of performance test you carry out will depend on the goals and scope of the non-functional requirements. A business needs to ensure that the key performance indicators (KPIs) for each performance feature can be met. This helps the business feel confident that the application or system being tested works as expected, without any bottlenecks, bugs or errors that could stop it from being released into production.

When running performance tests, it's important to meet these conditions first:

- The test environment should closely match the production environment (for example, in terms of RAM and the number of servers).
- There shouldn't be any unresolved severe or major defects.

Let's take a closer look at peak load testing.

[Skip to main content](#)

Peak load testing

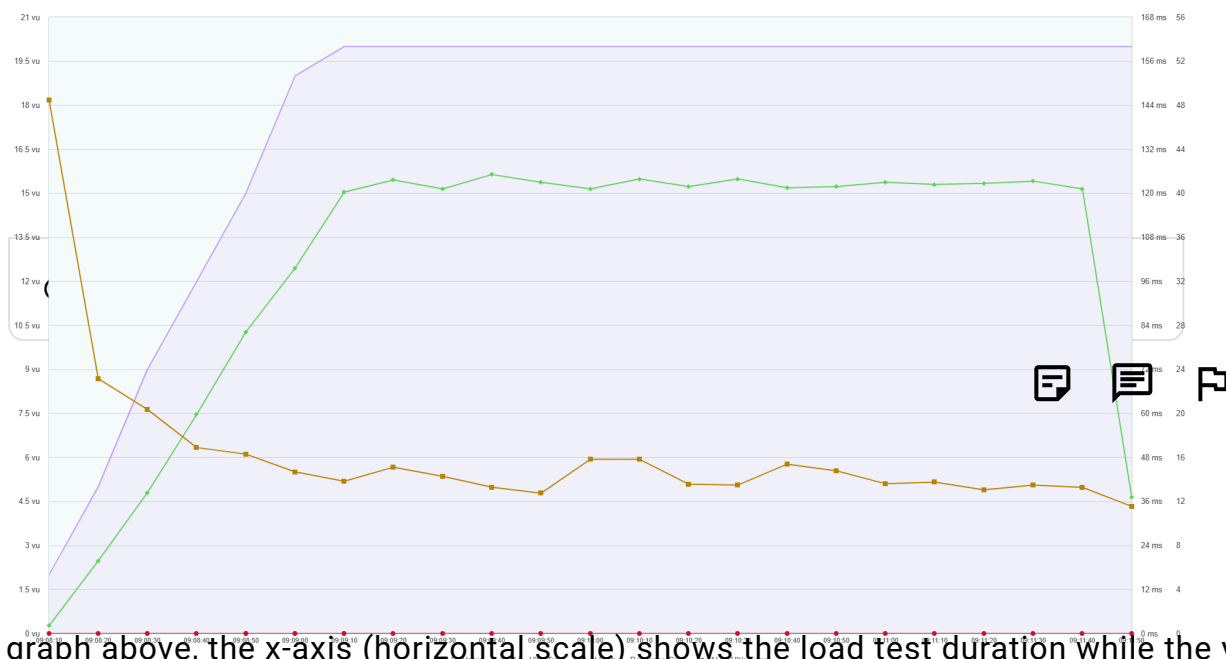
Note: Be cautious when attempting load tests on public servers. You can use the Apache benchmark tool (found in most LAMP/WAMP distributions as ab.exe) to run a load test. However, make sure to conduct the testing on your local web server to avoid triggering DoS (Denial of Service) attack alerts on public websites.

Peak load testing is a type of load testing that validates whether the software can perform under real-life load conditions. It enables you to estimate the processing speed for each request. Simulating high volumes of concurrent users performing requests or transactions over a long period of time will confirm how well the system handles these high volumes.

To do this, a performance test could be set up to add 100 users to an application every 20 seconds until reaching 1000 users (or the number specified in the nonfunctional requirements). The stepping process will take $100 \times 20 = 2000$ seconds. When it reaches 1000 threads, these will continue to run simultaneously for one hour. The results recorded will give you valuable information about your application's peak load capacity.



[Skip to main content](#)



In the graph above, the x-axis (horizontal scale) shows the load test duration while the y-axis (vertical scale) measures these four metrics:

The green line indicates the total request throughput. This means the number of calls or requests made to the server per second during the load test. In this case, the value is between 10 and 12 hits per second, approximately. This variable uses the y-axis scale on the far right of the graph.

The red line (which overlaps the x-axis) indicates the error throughput. This means the number of failed calls to the server per second during the test. Since there are zero errors, this line overlaps the x-axis. This variable uses the same scale as that on the far right of the graph.

The blue line indicates the number of (virtual) users accessing the system under test during this test. In this example, the test started with one user and slowly ramped up to 10 virtual users within the first minute of the test and remained constant at that level until the end of the test. This variable uses the y-axis scale on the left.

The orange line indicates the average response time. This is a crucial metric indicating how fast the system is performing. In this example, on average, the system under test is taking about 240 milliseconds to respond to the page requests made by the load-testing tool. Since the orange line is stable during the test, this means the system's performance is stable during the load test without any fluctuations. This variable uses the y-axis scale on the right of the graph.

[Skip to main content](#)

Overall, this report shows that the growing number of users doesn't affect the response time, the error rate continues to be low even as the throughput (hits per second) increases, and the performance is stable.

Stress testing

Stress testing gives you information about the upper limits of capacity within the system under test under extreme loads.

If stress testing involves a sudden ramp-up of virtual users, then it's known as a spike test. If it involves a slow ramp-up, it's called a soak test as this is testing the system's sustainability over time.

Unlike peak load testing, stress testing applies an extreme workload to the system. It aims to see how the system behaves under this extreme load, at what load it breaks and how well it recovers. Anticipated major events such as Black Friday or Boxing Day sales are better managed by executing performance testing, such as spike testing, prior to the event.

The following graphs show visual representations of a spike test and a soak test, respectively. Select the arrows to view each diagram.

[Prev](#)[Next](#)[Skip to main content](#)

Spike test graph



Designing the right type of test with the right data and variables is critical to the success of testing, particularly performance testing, where the load profile or parameters need to be appropriate for the performance criteria or test objectives.



☒ Performance test

Non-Assessed

[Skip to main content](#)

A retail website is launching a special deal for Black Friday. Black Friday is the most critical day in the e-commerce industry because retail websites experience an increase in traffic and revenue. A huge amount of traffic can increase the revenue of a retail store but when an e-commerce website is not ready to withstand the increase in traffic, it can result in the website crashing. How would you ensure that the website is ready for Black Friday and won't crash?

Answer the questions:

1. What do you need to do to ensure the retail website is ready for Black Friday?
2. What performance testing would you do and why?

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